

1 - 4. Find the derivative for each of the following Chain rule problems using the form  $\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$ . Show all work and answers using rigorous differential notation.

1. Let  $f(x) = (4x^3 - 2x^2 + 3)^5$ .  
Define  $f(u)$  and  $u(x)$ .

2. Let  $f(\theta) = (\sin(\theta))^5$ .  
Define  $f(u)$  and  $u(x)$ .

3. Let  $f(a) = (x(a))^5$ .

4. Let  $x(t) = (f(t))^3$ .

5. Solve for  $y$  in terms of  $x$ :  $y^2 - 1 = x^3$ . Can this result be expressed as  $y = f(x)$ ? Explain.

6. Solve for  $y$  in terms of  $x$ :  $y^5 - \sin(y) - 1 = x^2$ .

7. (a) Let  $y^3 - 1 = x^2$ . Find  $\frac{dy}{dx}$  by first solving for  $y$  as a function of  $x$ .

7. (b) Let  $y^3 - 1 = x^2$ . Do not solve for  $y(x)$  and use the Chain rule for implicit differentiation to find  $\frac{dy}{dx}$ .

#### The Chain rule of Implicit Differentiation

1. Rewrite the equation to show the explicit composite function.
2. Use the Chain rule to compute the derivative of the equation.
3. Write the derivative in shorthand notation.
4. Group derivative ( $y'$ ) terms on one side of the equal sign.
5. Isolate ( $y'$ ) by factoring and then solve for  $\frac{dy}{dx}$ .

(continued) 7.(b) Let  $y^3 - 1 = x^2$ . Although this equation can be solved for  $y$ , use the Chain rule for implicit differentiation.

- step 1: The quantity  $\frac{dy}{dx}$  *implies* that  $y$  MUST be a function of  $x$ . Rewrite the equation to show the explicit composite function.

- step 2: Find the derivative; use the Chain rule on the composite function in the equation in step 1.

- step 3: Rewrite step 2 using “shorthand” notation to make the algebraic process easier.

- step 4: Group all derivative ( $y'$ ) terms on one side and the non-derivative terms on the other.

- step 5: Isolate the derivative ( $y'$ ) by factoring and then solve for  $\frac{dy}{dx}$ .

8 - 10. Find  $\frac{dy}{dx}$  for each equation in 8 - 10. If the Chain rule for implicit differentiation is required, show the four-step method for implicit differentiation using rigorous differential notation.

8. Let  $y^4 + x + 3 = 6y$ .

(a) Can this equation can be solved for  $y$  in terms of only  $x$ ? Circle one.

Yes, so solve for  $y$  and find the derivative / No, so use the Chain rule for implicit functions

(b)

- step 1: The quantity  $\frac{dy}{dx}$  *implies* that  $y$  MUST be a function of  $x$ . Rewrite the equation to show the explicit function.

- step 2: Find the derivative; use the Chain rule on the composite function in the equation in step 1.

- step 3: Rewrite step 2 using "shorthand" notation to make the algebraic process easier.

- step 4: Group all derivative ( $y'$ ) terms on one side and the non-derivative terms on the other.

- step 5: Isolate the derivative ( $y'$ ) by factoring and then solve for  $\frac{dy}{dx}$ .

9. Let  $2y - 2x^3 + 3x^4 = 4 - x^4$ . Find  $\frac{dy}{dx}$ .

Can this equation can be solved for  $y$  in terms of only  $x$ ? Circle one.

Yes, so solve for  $y$  and find the derivative / No, so use the Chain rule for implicit functions

10. Let  $p^3 - 2x^3 + 3p^2 = x^2 - 4$ . Find  $\frac{dp}{dx}$ .

(a) Can this equation can be solved for  $p$  in terms of only  $x$ ? Circle one.

Yes, so solve for  $p$  and find the derivative / No, so use the Chain rule for implicit functions

(b)

- step 1: The quantity  $\frac{dp}{dx}$  *implies* that  $p$  MUST be a function of  $x$ . Rewrite the equation to show the explicit function.

- step 2: Use the Chain rule on the function equation from step 1:

- step 3: Rewrite step 2 using "shorthand" notation to make the algebraic process easier.

- step 4: Group all derivative ( $p'$ ) terms on one side and the non-derivative terms on the other.

- step 5: Isolate the derivative ( $p'$ ) by factoring and then solve for  $\frac{dp}{dx}$ .

11. Find the slope of the tangent line to the curve  $(x - 2)^3 - 2 \sin y = y - 8$  at the point  $(0,0)$ . Use the Chain rule for implicit differentiation.

Give the formula for the slope of the tangent line.

12. Use implicit differentiation to calculate  $\frac{dy}{dx}$  from the equation  $x^4 + y^4 = 2$ , and find the equation of the tangent line to the curve at the point  $(1, -1)$ . Show all derivative work using rigorous differential form and give the formulas that are used in the problem.

**Recall:**  $x = \log_b a$  if and only if  $b^x = a$

1.  $a > 0$  and  $b > 0$  so only positive values allowed and  $b \neq 1$
2.  $b$  is the “base” and  $a$  is called the “argument” of the logarithm;
3.  $x$  is the exponent or power

“Logs roll”:  $\log_{\text{base}} x = \text{power} \rightarrow \text{base}^{\text{power}} = x$

Zero rule for logarithms:  $\log_b 1 = 0$

Simplify each of the following.

(a)  $\log 1 =$

(b)  $\ln 1 =$

(c)  $\log_5 1 =$

Rule of 1 for logarithms:  $\log_b b = 1$

Simplify each of the following.

(a)  $\log_7 7 =$

(b)  $\ln e =$

(c)  $\log 10 =$

Product to sum / Sum to product rule:

$$\log_b (rs) = \log_b r + \log_b s$$

1. Use the properties of logarithms to simplify the following expressions such that the argument of the logarithm does not contain any product.

(a)  $\log_x abc$

domain restriction:

(b)  $\ln (x - y)$

domain restriction:

(c)  $\log 10xy$

domain restriction:

(continued) 1. Use the properties of logarithms to simplify the following expressions such that the argument of the logarithm does not contain any product.

$$(d) \log_B [2a(x + y)]$$

domain restriction:

2. Write the expression as a single logarithm with a coefficient of 1.

$$\log_3 x + \log_3 (3 + y) + \log_3 z$$

domain restriction:

Quotient to difference / difference to quotient rule:

$$\log_b \frac{r}{s} = \log_b r - \log_b s$$

3. Use the properties of logarithms to simplify the following expressions such that the argument of the logarithm does not contain any product or quotient.

$$(a) \log \frac{(x-y)}{ab}$$

domain restriction:

$$(b) \log_2 \frac{xy}{2m}$$

domain restriction:

4. Write the expression as a single logarithm with a coefficient of 1.

$$\ln a - \ln y + \ln w$$



Power rule: Powers roll over logs:  $\log_b r^s = s \log_b r$

5. Use the properties of logarithms to simplify the following expressions and functions so that the argument of the logarithm does not contain any product, quotient or power.

(a)  $\ln x^5$

(b)  $\log xy^7$

First, explain why the power rule does not apply directly to the expression  $\log xy^7$ ; then simplify.

(c)  $\log_5 a^5 b^3 c^9 =$

domain restriction:

(d)  $\log_2 \frac{x^2 y^4}{2m^3} =$

6. Write the expression as a single logarithm with a coefficient of 1.

$4 \log_b w - 2 \log_b w + \log_b x$

domain restriction:

Inverse relationship for exponential and logarithmic functions:  $b^{\log_b x} = x$  and  $\log_b b^x = x$ .

$4^{\log_4 x}$

$10^{2 \log 7}$

Change of Base Formula:

Use the change of base formula to rewrite each of the following functions and expressions. Simplify completely.

1.  $f(x) = \log_x 2$

2.  $k(x) = \log_3 15$

3.  $g(x) = \log_7 e$

4.  $k(y) = \frac{\ln y}{\ln 2}$

5.  $f(z) = \log_3(z - 4)$

6.  $p(x) = \frac{\log x}{\log 5}$

7.  $q(x) = \frac{\log_2 7}{\log_2 x}$

## Summary and examples

For positive real numbers  $r$ ,  $s$ , and  $x$ , and for a real number  $b > 0$  and  $b \neq 1$ :

**RULE****EXAMPLES**

1.  $\log_b 1 = 0$   
Rule of zero

$$\log_6 1 = 0$$

$$\log 1 = 0$$

$$\log_{71} 1 = 0$$

2.  $\log_b b = 1$   
Rule of one

$$\log 10 = 1$$

$$\log_6 6 = 1$$

$$\log_{23} 23 = 1$$

3.  $\log_b (rs) = \log_b r + \log_b s$   
Product to sum / Sum to product

$$\ln (22) = \ln 2 + \ln 11$$

$$(\text{Recall: } \ln 22 = \log_e 22)$$

$$\log_3 (70) = \log_3 (7 \cdot 5 \cdot 2) = \log_3 7 + \log_3 5 + \log_3 2$$

$$\log (2xy) = \log 2 + \log x + \log y$$

$$\log_6 5 + \log_6 a + \log_6 b = \log_6 (5ab)$$

4.  $\log_b \frac{r}{s} = \log_b r - \log_b s$   
Quotient to difference / difference to quotient

$$\log \frac{x}{2} = \log x - \log 2$$

(Note the +/- signs)

$$\ln \frac{ax}{y} = \ln a - \ln y + \ln x$$

(Note the +/- signs)

$$\log_7 w - \log_7 b - \log_7 x = \log_7 \frac{w}{bx}$$

(Note the +/- signs)

5.  $\log_b r^s = s \log_b r$   
Power rule: Powers roll over

$$\ln 7^{25} = 25 \ln 7$$

$$\log_3 16 = \log_3 2^4 = 4 \log_3 2$$

$$\log_7 200 = \frac{\ln (8 \cdot 25)}{\ln 7} = \frac{\ln 8 + \ln 25}{\ln 7} = \frac{3 \ln 2 + 2 \ln 5}{\ln 7}$$

$$\hookrightarrow 8 = 2^3 \hookrightarrow 25 = 5^2$$

$$\ln \frac{x^2}{y^3 z^4} = 2 \ln x - 3 \ln y - 4 \ln z$$

$$\frac{1}{2} \ln a - 2 \ln b + 4 \ln c = \ln \frac{\sqrt{a} c^4}{b^2}$$

6.  $b^{\log_b x} = x$   
Logarithms as Exponents

$$10^{\log x} = x$$

$$e^{\ln x} = x$$

$$4^{3 \log_4 x} = x^3$$

### Absolute Value Function

Recall the definition of the absolute value:  $f(x) = |x| = \sqrt{x^2} = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$ .

For  $x < 0$ :  $|x| = -(x)$

example:  $|-5| = -(-5) = 5$

For  $x \geq 0$ :  $|x| = x$

example:  $|5| = 5$

So, when  $x$  is negative we must insert an additional negative sign to insure that the output (function) always generates a positive value when we “remove” the absolute value symbol.

When  $x$  is positive we don’t need to insert a negative sign since  $f(x) = |x| = (x)$  will generate an output (function) that is always a positive value when we “remove” the absolute value symbol.

### Logarithmic functions

1. Consider the logarithmic function  $x = \log_b a$ . What are the restrictions on the values for  $a$  and  $b$ ?

2. Now consider the function  $f(x) = \ln x$ .

(a) What is the base for this logarithmic function?  
Does it satisfy the restrictions given in #1?

(b) What is the domain for  $f(x) = \ln x$ ?

3. Now consider the function  $f(x) = \ln |x|$ .

What is the domain for  $f(x) = \ln |x|$ ?

4. Use the definition of the absolute value function to give the piecewise definition of  $f(x) = \ln |x|$ .

Derivatives of Logarithmic functions

1. Consider the function  $y = f(x) = \ln |g(x)|$ .

(a) Convert the logarithmic equation into the equivalent exponential equation.

(b) The Chain rule for implicit differentiation is used to find the derivative  $\frac{dy}{dx}$  formula for the function  $y = \ln |g(x)|$ . However, in general the absolute value function can't be used when calculating a derivative. Why not? (Hint: consider the graph of an absolute value function and the conditions for the existence of a derivative.)

(c) Use the Chain rule for implicit differentiation to find the formula for the derivative  $\frac{dy}{dx}$  when  $y = \ln |g(x)|$ .

Recap:

Derivatives of Exponential and Logarithmic functions

1.

$$\frac{d}{dx} (\ln |g(x)|) =$$

$$\frac{d}{dx} (\ln g(x)) =$$

$$\frac{d}{dx} (\ln |x|) =$$

$$\frac{d}{dx} (\ln x) =$$

2.. Use the change in base formula to find the following derivatives for logarithms using a base  $b$  .

$$\frac{d}{dx} (\log_b |g(x)|) =$$

$$\frac{d}{dx} (\log_b g(x)) =$$

$$\frac{d}{dx} (\log_b |x|) =$$

$$\frac{d}{dx} (\log_b x) =$$

3.

$$\frac{d}{dx} (e^x) =$$

$$\frac{d}{dx} (e^{g(x)}) =$$

$$\frac{d}{dx} (b^x) =$$

$$\frac{d}{dx} (b^{g(x)}) =$$

Find each of the following derivatives. Give only the completely simplified results in the boxes. Below each box, give the domain of the function where indicated.

Template:

$$y' = \left( \frac{1}{\text{unchanged inside function}} \right) \left( \frac{\text{derivative of the inside function}}{1} \right) = \left( -\frac{1}{u} \right) (u')$$

1. Let  $y = \ln 4x^5$ . Find  $\frac{dy}{dx}$  and its domain.

2. Let  $f(x) = \ln (x^3 + 2x + 1)$ . Find  $f'(x)$ .

3. Let  $g(x) = 3 \ln |1 - x^2|$ . Find the derivative and its domain.

4. Let  $k(x) = 4 \ln (e^{2x} - 3 \cos x)$ . Find  $\frac{dk}{dx}$ .

5. Find  $\frac{dy}{dx}$  for  $y = 5 \log_4 (2x^2 - 3)$ .

6. Let  $y(x) = 4 \ln \sqrt[5]{x^2}$ . Find  $y'$  and its domain.

7. Let  $f(x) = (\ln x)^4$ . Find  $f'(x)$  and its domain.



8. Let  $k(a) = \log_2 (5a^3 - \sec a)$ . Find  $\frac{dk}{da}$ .

9 - 12. Find the first order derivative for each of the following functions.

9.  $y = 5x \ln (2x^2)$

10.  $y = \frac{\ln x}{x^4}$

(continued) 9 - 12. Find the first order derivative for each of the following functions.

11.  $f(x) = (e^{6x} \ln x^4)^3$

12.  $y = \ln\left(\frac{x+1}{3x-1}\right)$

13. Suppose that the population of a certain colony of honey bees is given by the function

$$P(t) = (t + 100) \ln(t + 2)$$

where  $t$  represents the time in days. Find the instantaneous rate of change in the population after 4 days. Is the population increasing or decreasing?

14. Let  $f(x) = \ln(3x - \sin(\pi x))$ . Find the equation of the tangent line to the graph of  $f(x)$  at  $x = 1$ .

15. Use logarithmic differentiation to find the derivatives of  $f(x) = b^x$  and  $g(x) = b^{-x}$ , where  $b \neq e$

$$\frac{d}{dx}(b^x) =$$

$$\frac{d}{dx}(b^{g(x)}) =$$

16. Let  $y = 4^x$ . Find  $\frac{dy}{dx}$ .

17. Let  $f(x) = 3^{x^3+2x+1}$ . Find  $\frac{df}{dx}$ .

18. Let  $g(x) = 5^{\sec x}$ . Find  $\frac{dg}{dx}$ .

The Chain rule of Implicit Differentiation

1. Rewrite the equation to show the explicit composite function.
2. Use the Chain rule to compute the derivative of the equation.  
Check to determine if the product or quotient rule should be used.
3. Write the derivative in shorthand notation.
4. Group derivative ( $y'$ ) terms on one side of the equal sign.
5. Isolate ( $y'$ ) by factoring and then solve for  $\frac{dy}{dx}$ .

1-3. Find  $\frac{dy}{dx}$  for each equation in 8 - 10. If the Chain rule for implicit differentiation is required, show the four-step method for implicit differentiation using rigorous differential notation.

1. Find  $\frac{dy}{dx}$  when  $y = xe^y$ . Can we solve for  $y$  as a function of  $x$ ? Yes / No

(a) Rewrite the equation to show the composite function explicitly. How do you know that the product rule must be used?

(b) Find the derivative.

2. Find  $\frac{dy}{dx}$  when  $2xy + y^2 = 6x$ . Simplify completely.  
Can we solve for  $y$  as a function of  $x$ ? Yes / No

3. Let  $(x + 2y)^3 = x + y^2 + 6$  Find  $\frac{dy}{dx}$ .  
Simplify completely.

Can we solve for  $y$  as a function of  $x$ ? Yes / No

4. Let  $(x + 2p) \sin p = 6x$  Find  $\frac{dp}{dx}$ . Can we solve for  $y$  as a function of  $x$ ? Yes / No.

5. Determine the equation of the tangent line to the curve  $x^2y^3 - xy = 30$  at the point  $(2,3)$ . Show all work in correct mathematical form, including formulas.

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6. Find the slope of the tangent line to the curve  $\cos(x - y) + \sin y = \sqrt{2}$  at the point  $\left(\frac{\pi}{2}, \frac{\pi}{4}\right)$ .

7. Determine the slope of the tangent line to the curve  $\frac{x-1}{y^2} + \frac{1}{y} = 6$  at  $\left(2, -\frac{1}{3}\right)$ . Hint: the derivative of the first term requires the use of the quotient rule.

Implicit Differentiation and Related Rates

1. Suppose that the two variables  $x$  and  $y$  are both functions of time,  $x(t)$  and  $y(t)$ , and  $x$  and  $y$  are related by the equation  $y^2 + x^3 = 6 - x$ . When  $(x,y) = (4, -1)$ ,  $\frac{dx}{dt} = -3$ . Find  $\frac{dy}{dt}$ .

- step 1: Rewrite the equation to show the explicit functions.

- step 2: Use the Chain rule on the function equation from step 1:

- step 3: Substitute the values for  $x$ ,  $y$ , and  $\frac{dx}{dt}$ .

- step 4: Solve for  $\frac{dy}{dt}$ .

2. The quantity  $x$  demanded weekly of a particular product is related to its unit price  $p$  by the equation  $p^2 + 2x^4 = 57$ , where  $p$  is measured in dollars and  $x$  is measured in units of a hundred. How fast is the quantity demanded  $x$  changing when  $x = 2$  and  $p = 5$  and the price per unit is increasing at a rate of \$2 per week? Give an interpretation of the answer.

- step 1: Rewrite the equation to show the explicit functions.

- step 2: Use the Chain rule on the function equation from step 1:

- step 3: Define all numerical values from the problem statement using the correct variable and derivative notation. Include units for all quantities. State the derivative that the problem requires us to find in correct derivative notation.

- step 4: Substitute all numerical values into the derivative equation.

- step 5: Solve for the unknown derivative.

3. The sides of a square *decrease* in length at a rate of 1 m/s. At what rate is the area of the square changing when the sides are 5m long? Give an interpretation of the answer.

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- Step 1: Draw a sketch and label the variable of interest. Next, write an equation that represents the quantity that is required to find the solution to the problem

- step 2: Rewrite the equation in step 1 to show the explicit functions.

- step 3: Use the Chain rule on the function equation.

- step 4: Define all numerical values from the problem statement using the correct variable and derivative notation. Give the derivative that the problem requires us to find in correct derivative notation.

- step 5: Substitute all numerical values into the derivative equation.

- step 6: Solve for the unknown derivative.

4. The volume of a cylinder is given by  $V = \pi r^2 h$ . The radius,  $r$ , of the cylinder is a constant 2 cm and the volume of the cylinder is increasing at a rate of  $5 \text{ cm}^3/\text{s}$  when the height,  $h$ , is 10 cm. At what rate is the height,  $h$ , changing?

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- step 1: Rewrite the equation to show the explicit functions.

- step 2: Use the Chain rule on the function equation from step 1:

- step 3: Define all numerical values from the problem statement using the correct variable and derivative notation. Include units for all quantities. State the derivative that the problem requires us to find in correct derivative notation.

- step 4: Substitute all numerical values into the derivative equation.

- step 5: Solve for the unknown derivative.

5. A ladder 10 ft long is leaning against a wall. The bottom of the ladder is a distance  $x$  from the wall and the top of the ladder is a distance  $y$  from the ground. The relationship between  $x$  and  $y$  is given by  $x^2 + y^2 = 100$ .

When the bottom of the ladder is 8 ft from the wall, the bottom of the ladder is sliding away from the wall at a rate of 4 ft/sec. Find the rate at which the top of the ladder is sliding down the wall. Show all 5 steps for the solution and include a sketch for the problem.